

Task 1: Create mysite Directory and Secure It

Steps:

- Go to your home directory
 - `cd /home`
- Create the website directory
 - `mkdir mysite`
- Create the web page
 - `vim mysite/index.html`
 - Add the following HTML:

```
<!DOCTYPE html>
<html>
<head>
<title>My Secure Site</title>
</head>
<body>
<h1>Welcome to My Secure Website</h1>
<p>This site is accessible only to the owner.</p>
</body>
</html>
```
- Change ownership
 - `chown -R yourusername:yourusername /home/mysite`
- Set permissions
 - `chmod 700 /home/mysite`
 - `chmod 600 /home/mysite/index.html`
- Verify:
 - `ls -ld /home/mysite`
 - `ls -l /home/mysite`

```
root@mail/home
[nishitpanchal@mail ~]$ sudo -i
[sudo] password for nishitpanchal:
Sorry, try again.
[sudo] password for nishitpanchal:
[root@mail ~]# cd /home
[root@mail home]# mkdir mysite
[root@mail home]# vim mysite/intex.html
[root@mail home]# chown -R nishitpanchal:nishitpanchal /home/mysite
[root@mail home]# chod 700 /home/mysite
bash: chod: command not found...
^[[A[root@mail home]# chod 700 /home/mysite
bash: chod: command not found...
[root@mail home]# chmod 700 /home/mysite
[root@mail home]# chmod 600 /home/mysite/intex.html
[root@mail home]# ls -ld /home/mysite
drwx-----. 2 nishitpanchal nishitpanchal 24 Jul 11 23:05 /home/mysite
[root@mail home]# ls -l /home/mysite
total 4
-rw-----. 1 nishitpanchal nishitpanchal 190 Jul 11 23:05 intex.html
[root@mail home]#
```

Task 2: Configure Apache

Start and enable Apache

- `systemctl start --now httpd`
- `systemctl enable --now httpd`

Check status

- `systemctl status httpd`

Allow HTTP through firewall

- `firewall-cmd --permanent --add-service=http`
- `firewall-cmd --reload`

```
Activities Terminal Jul 11 23:12 root@mail:~

[root@mail ~]# systemctl start --now httpd
[root@mail ~]# systemctl enable --now httpd
[root@mail ~]# systemctl status --now httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: >
   Active: active (running) since Sat 2026-07-11 22:56:16 IST; 14min ago
     Docs: man:httpd.service(8)
  Main PID: 1008 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Byt>
    Tasks: 177 (limit: 10524)
   Memory: 13.8M (peak: 18.7M)
      CPU: 1.120s
    CGroup: /system.slice/httpd.service
            └─1008 /usr/sbin/httpd -DFOREGROUND
              └─2780 /usr/sbin/httpd -DFOREGROUND
                └─2784 /usr/sbin/httpd -DFOREGROUND
                  └─2786 /usr/sbin/httpd -DFOREGROUND
                    └─2789 /usr/sbin/httpd -DFOREGROUND

Jul 11 22:55:53 mail.example.com systemd[1]: Starting The Apache HTTP Server.>
Jul 11 22:56:16 mail.example.com systemd[1]: Started The Apache HTTP Server.
Jul 11 22:56:16 mail.example.com httpd[1008]: Server configured, listening on>
[root@mail ~]# firewall-cmd --add-service=http --permanent
usage: 'firewall-cmd --help' for usage information or see firewall-cmd(1) man page
firewall-cmd: error: unrecognized arguments: --add-service=http
[root@mail ~]# firewall-cmd --reload
success
[root@mail ~]# firewall-cmd --list-all
public (active)
  target: default
 icmp-block-inversion: no
 interfaces: ens32
  sources:
 services: cockpit dhcpv6-client http smtp ssh
  ports:
 protocols:
 forward: yes
 masquerade: no
 forward-ports:
 source-ports:
 icmp-blocks:
```

Configure Apache

Open configuration file

- `vim /etc/httpd/conf.d/mysite.conf`

Add:

```
<VirtualHost *:80>
  DocumentRoot "/home/mysite"
  <Directory "/home/mysite">
    Require all granted
  </Directory>
</VirtualHost>
```



```
root@mail~  
[root@mail ~]# vim /etc/httpd/conf.d/mysite.conf  
[root@mail ~]# restorecon -Rn /home/mysite  
[root@mail ~]# systemctl restart httpd  
[root@mail ~]# ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:47:5c:f6 brd ff:ff:ff:ff:ff:ff  
    altname enp2s0  
    inet 192.168.182.136/24 brd 192.168.182.255 scope global dynamic noprefixroute ens32  
        valid_lft 1287sec preferred_lft 1287sec  
    inet6 fe80::20c:29ff:fe47:5cf6/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[root@mail ~]# ifconfig  
ens32: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.182.136 netmask 255.255.255.0 broadcast 192.168.182.255  
    inet6 fe80::20c:29ff:fe47:5cf6 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:47:5c:f6 txqueuelen 1000 (Ethernet)  
    RX packets 68316 bytes 93635608 (89.2 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 42143 bytes 3279978 (3.1 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 62 bytes 30098 (29.3 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 62 bytes 30098 (29.3 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
[root@mail ~]# S
```

Task 3: Restrict Another User

Create another user

- useradd testuser
- passwd testuser

Switch user

- su - testuser

Try accessing the directory

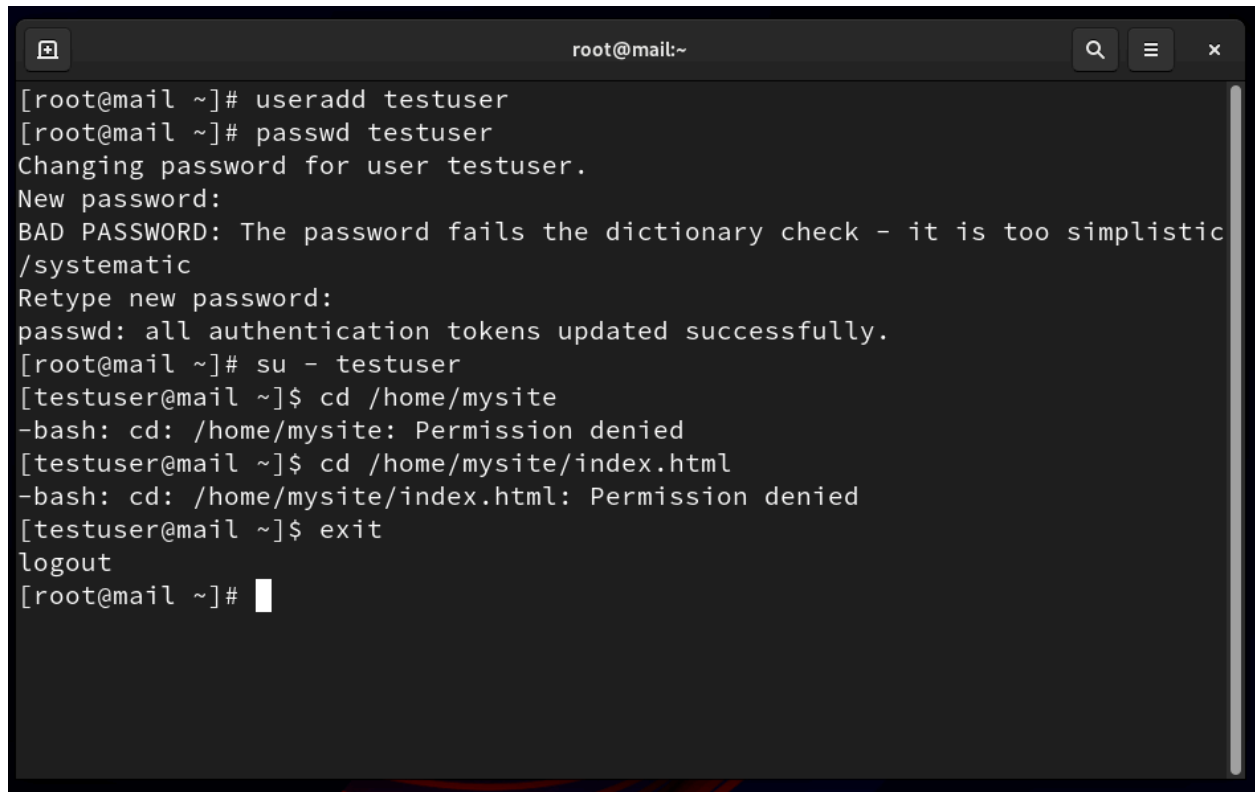
- cd /home/mysite

Try reading the file

- cat /home/mysite/index.html

Exit

- exit

A terminal window titled 'root@mail:~' with search, menu, and close icons in the title bar. The terminal shows the following commands and output:

```
[root@mail ~]# useradd testuser
[root@mail ~]# passwd testuser
Changing password for user testuser.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic
/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@mail ~]# su - testuser
[testuser@mail ~]$ cd /home/mysite
-bash: cd: /home/mysite: Permission denied
[testuser@mail ~]$ cd /home/mysite/index.html
-bash: cd: /home/mysite/index.html: Permission denied
[testuser@mail ~]$ exit
logout
[root@mail ~]#
```

Task 4: Documentation

Commands Used

- `mkdir /home/mysite`
- `vim /home/mysite/index.html`
- `chown -R yourusername:yourusername /home/mysite`
- `chmod 700 /home/mysite`
- `chmod 600 /home/mysite/index.html`
- `yum -y install httpd`
- `systemctl enable --now httpd`
- `firewall-cmd --permanent --add-service=http`
- `firewall-cmd --reload`
- `restorecon -Rv /home/mysite`
- `setsebool -P httpd_enable_homedirs on`
- `systemctl restart httpd`
- `hostname -l`
- `useradd testuser`
- `passwd testuser`

- su - testuser
- cd /home/mysite
- cat /home/mysite/index.html

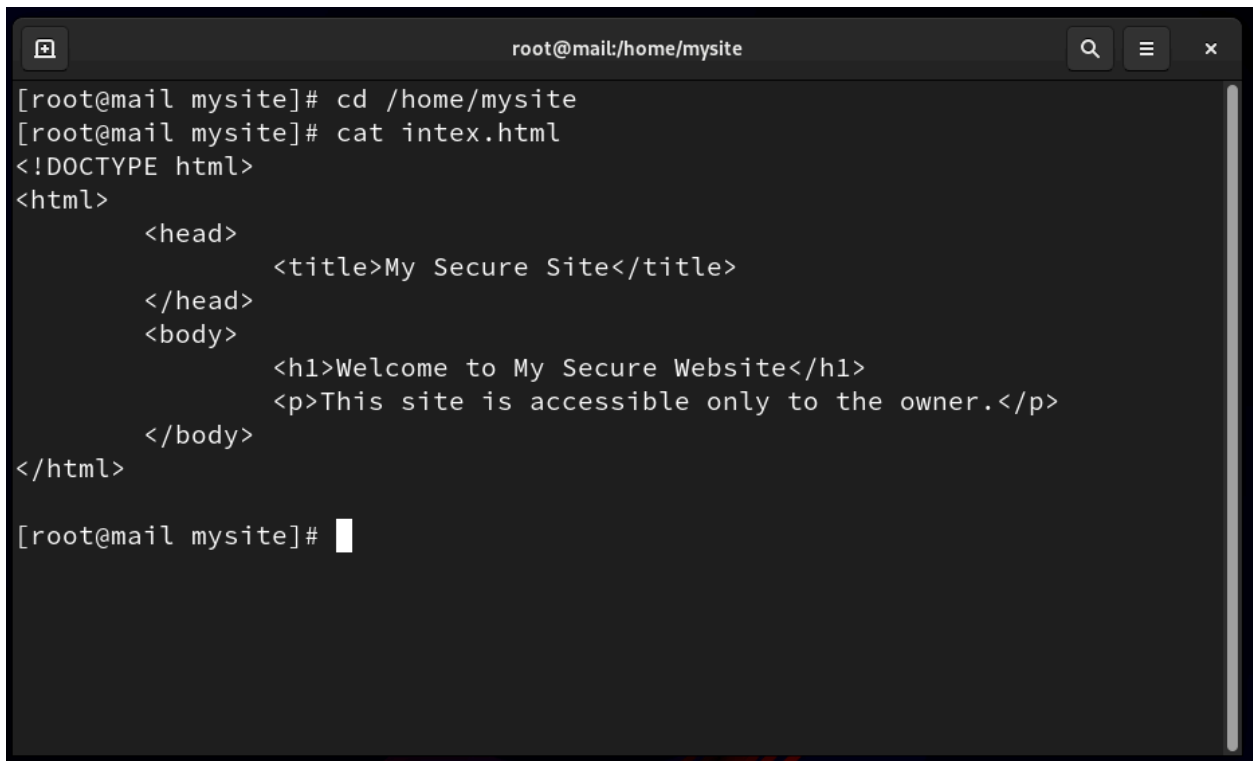
Permission Settings

File/Directo ry	Permissi on	Meaning
/home/mysite	700	Only owner has full access
index.html	600	Only owner can read and modify

Expected Results

Owner

- cd /home/mysite
- cat index.html

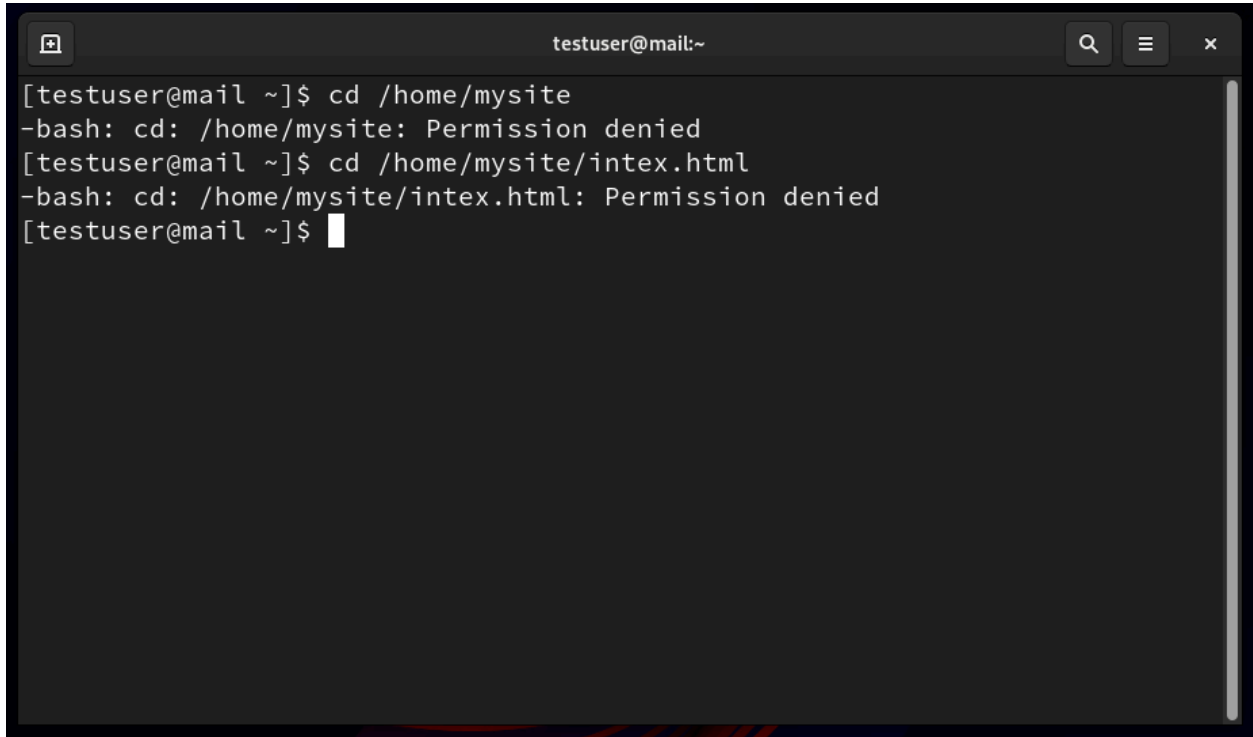


```
root@mail:/home/mysite
[root@mail mysite]# cd /home/mysite
[root@mail mysite]# cat index.html
<!DOCTYPE html>
<html>
  <head>
    <title>My Secure Site</title>
  </head>
  <body>
    <h1>Welcome to My Secure Website</h1>
    <p>This site is accessible only to the owner.</p>
  </body>
</html>

[root@mail mysite]#
```

Another User

- `cd /home/mysite`
- `cd /home/mysite/index.html`

A terminal window titled "testuser@mail:~" with search, menu, and close buttons in the title bar. The terminal shows a user attempting to change directories but being denied permission.

```
[testuser@mail ~]$ cd /home/mysite
-bash: cd: /home/mysite: Permission denied
[testuser@mail ~]$ cd /home/mysite/intex.html
-bash: cd: /home/mysite/intex.html: Permission denied
[testuser@mail ~]$
```